

Terahertz sensors for next generation biomedical and other industrial electronics applications: A critical review

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ABSTRACT

Terahertz (THz) sensors have emerged in recent years as an innovative technique for studying biological and chemical materials in a non-destructive, label-free and non-contact manner. The interaction between analyte and THz wave is very weak when detecting analytes such as DNA/RNA, proteins and other tiny biomolecules because the wavelength of THz wave is large and does not correspond to the scale of analyte. Therefore, there is an immediate requirement to look into a novel, extremely sensitive, and specialized biosensing technique appropriate for the THz band in order to detect trace analytes accurately. This article critically reviews the significance of THz-sensors for biosensing, chemical sensing, disease detection, dielectric & refractive index (RI) sensing, temperature sensing and strain sensing. Split ring resonators (SRR), metamaterial absorbers (MMA), plasmonic gratings, Fano-resonators, photonic crystal resonators (PCR), meta surfaces and plasmonic waveguides are the popular sensing structures that can be adopted in the development of THz-sensors. Quality factor (Q-factor), sensitivity and figure of merit (FoM) are the key factors which can be used for measuring the performance of THz-sensors with both transmissive or reflective mode of operation.

1. Introduction

The demand for innovative biological and chemical sensing technologies that operate in the infrared, optical, microwave and terahertz (THz) regimes has been substantially growing. Biochemical molecules, in particular, exhibit distinctive fingerprint characteristics in the THz regime due to the intra and inter-molecular variations that take place in this regime. The THz technology offers unique benefits such as non-destructive, non-ionizing and strong penetration for biochemical sensing [1–4]. A great deal of recent research in this sector have been devoted to improving the sensitivity of THz sensors. In an effort to boost sensitivity when using THz sensing, several artificial material techniques have been put forth including photonic crystals [5], split ring resonators (SRR) [6], meta-surfaces and parallel plate waveguides [7], [8]. The fundamental idea behind these techniques is to adjust the sensor structure's geometrical parameters to bring the resonant frequency (fr) to a desirable position where it matches the characteristic frequency of the analyte. As a result, the geometrical features of the THz Sensor structure

must usually be modified to sense diverse analytes. The earlier metamaterials were intended to be easily fabricable planar structures. The SRR is a type of powerful artificial structure exhibiting metamaterial behaviour that is made up of two concentric open loops with opposing openings in each SRR. Fano-resonators are very complex planar structures. The complex structured SRRs and Fano resonators offer high Q-factor and improved sensitivity compared to simple planar metamaterials. Compared with other resonators, all dielectric PCR may be adopted to improve wave confinement and decrease radiation losses. Nevertheless, there is a large path loss in the free-space measurement and a high level alignment is needed for PCR. A sort of primary excitation in precious metals like gold called SPP (surface plasmon polariton) can contribute to the electromagnetic waves (EMWs) resonating with the charge oscillation on the material's surface. This SPP phenomenon can be used to develop SPP-based THz-sensors. Recent investigations have demonstrated that SPP exists for graphene and their micro - nano structures has become a rising star in the development of THz-sensors. The SPR process involves stimulating the resonant

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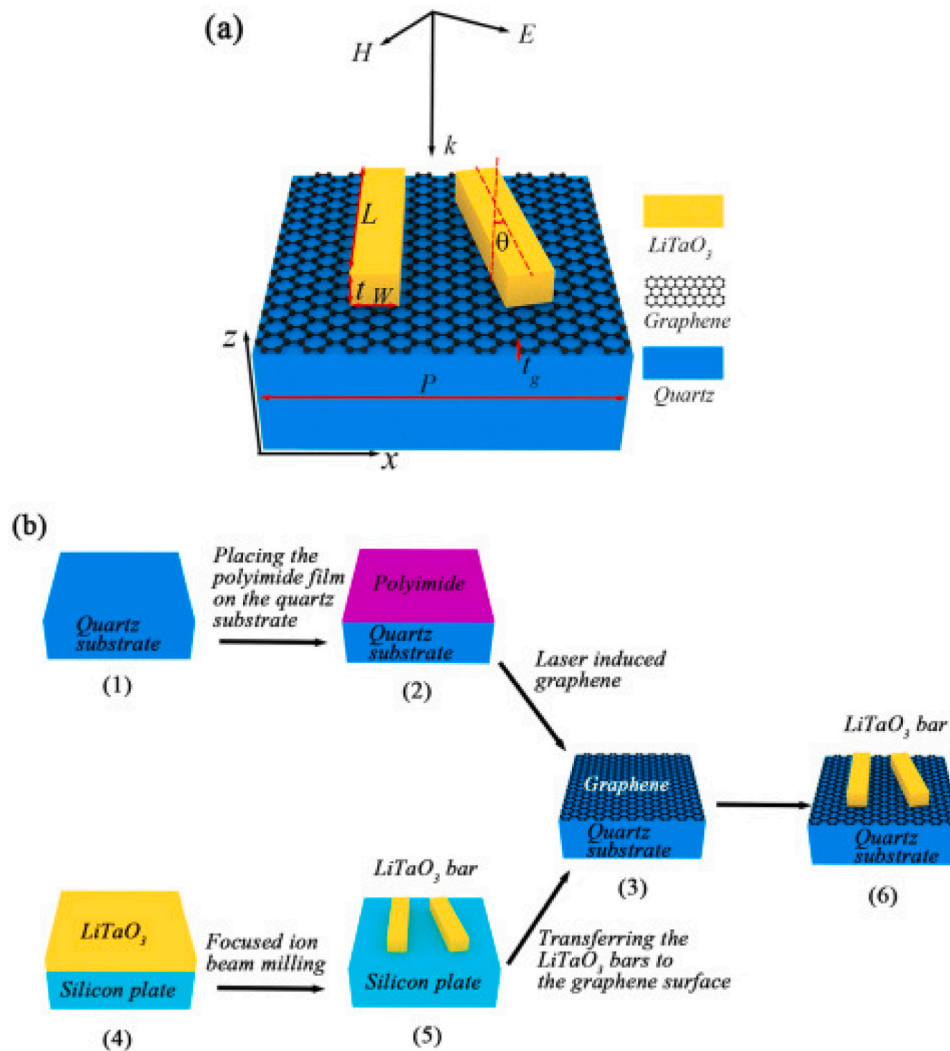


Fig. 1. (a) GDHM unit cell (b) fabrication steps for GDHM (Reprinted from [16] with permission from Elsevier).

oscillation of free electrons at the interface of materials with positive and negative permittivity using incident THz waves. Since the SPW (surface plasmon waves) are located at the boundary, SPRs are extremely sensitive to any change in the boundary which can be utilized to develop THz-sensors. SPR-based THz-sensors have large demand in diverse fields such as biochemical test, medical and analysis, environmental engineering and food safety monitoring.

The combination of THz sensing technology and meta surfaces has the benefits of being label free as well as having high sensitivity and it can improve the interaction between electromagnetic wave (EMW) and analyte [9]. The advent of diverse synthetic materials increasing the possibility of detecting exceedingly sensitive biomolecular fingerprints using THz spectroscopy. A sort of sub wavelength artificially made periodic material known as THz meta material has demonstrated potential for use in the THz spectrum, thanks to its intriguing novel electromagnetic characteristics, including perfect absorption, a negative refractive index (RI) and a negative dielectric constant [10–16]. Meta materials (consisting of 2D & 3D meta surfaces) can significantly enhance the sensitivity of THz-sensors. Characteristics of meta materials can be achieved by implementing SRRs. GDHM (graphene-dielectric hybrid meta material) (Fig. 1) is an emerging material which helps to enhance the sensitivity of THz-sensors [16]. In recent years, SRRs have been established to be employed in the development of THz biosensors. SRRs in THz- sensors can be fabricated with diverse configurations including

square for circular shapes with multiple or single gaps. In recent years, THz-sensors have gained tremendous popularity in a wide range of applications, including soft robotics, security applications, food Industry, environmental monitoring, healthcare and so on. THz-sensors can be also used for gas sensing [11–16], strain sensing [17–20], automobile paint quality sensing [21], sensing of chiral substances [22], imaging and security [23–26], indoor aerosol sensing [27], sensing ultra-fast switching in Marx generators [28], vibration sensing [29], sensing of concealed objects [30], precision sensing [31], characterization of liquid analytes [32] and trace isomer sensing [33].

THz-based sensors provide intriguing advantages for quality inspection, particularly when THz technology is built on all fiber photo-conductive technology that combines compact and resilient deployment with a wide dynamic range [34–46]. THz- imaging with the help of THz-sensors can be used to for quality control. THz- technology foundation of the speed and data hungry future generation communication, spectroscopy, security and imaging. Future generations will need linear, highly sensitive and broadband THz-sensors [38–42]. The THz-technology provides a spectrum window with numerous prospects for science, but its use is currently restricted. One of the primary barriers impeding the development of THz technology in the future is the lack of highly sensitive THz detectors/sensors [43]. This article critically reviews the performance and challenges in the development of THz-sensors for future industrial applications.

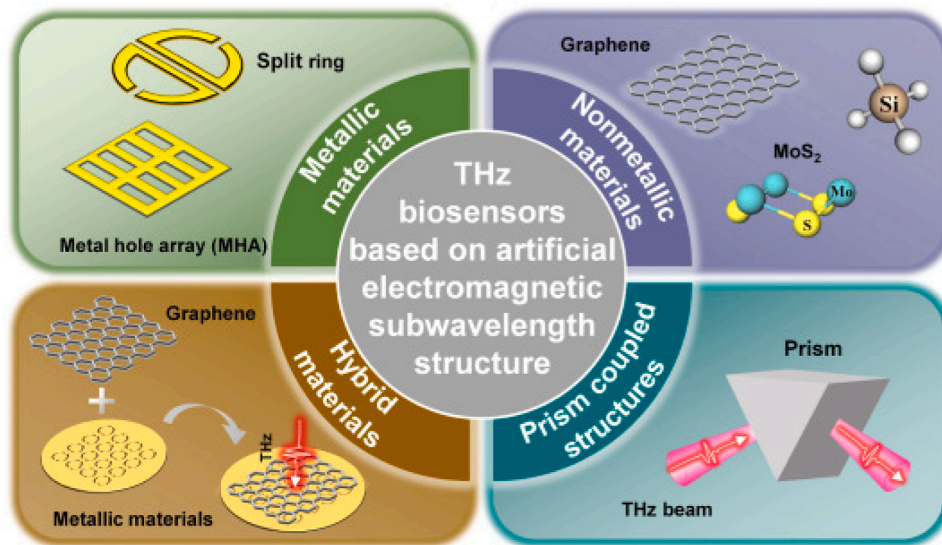


Fig. 2. THz-biosensors with various materials and architectures (Reprinted from [47] with permission from Elsevier).

2. Terahertz sensors for biosensing

In the electromagnetic spectrum THz-waves (0.1–10 THz)/ (30–3000 μm) lies between mid infrared and microwaves regimes [47–52]. THz-waves are highly promising for developing biosensors due to their unique features like no ionization hazards to biomolecules, low Photon energy level and low emissivity. THz-biosensors can be used for detecting biomolecules such as proteins, nucleic acids, enzymes, blood cells, DNA/RNA, pesticides, tissues and so on. The diverse types of THz-biosensors with various materials and architectures are displayed in Fig. 2. In 2016, F. Zangeneh et al., [53] reported an innovative label free THz- ring resonator biosensor that consisted of Si₃N₄ dielectric ring resonator vertically attached to a thin graphene strip ring resonator, porous-Al₂O₃ cladding on top of the graphene strip to improve the interaction between biomolecules and surface plasmon (SP) waves and SiO₂ buffer layer. The μ_n (chemical potential) of graphene significantly influences the Q-factor, sensitivity and FoM of this sensor. The increase in μ_n causes the increase of Q-factor and decrease of sensitivity. The optimum μ_n was about 0.6 eV.

In 2017, Ibraheem Al-Naib et al., [54] reported a label free THz meta material resonator biosensor that consisted of Si-substrate and Au-based joint split ring resonators. This sensor works based on the shifting of fr with respect to variation in RI of the analyte. BSA (Bovine Serum Albumin) is the most available protein in blood. As a result, observing how BSA interacts with drugs in aqueous form is particularly beneficial for understanding how drugs respond in the body and so provides specific guidance for medication administration and therapy. In 2017, Fangrong Hu et al., [55] reported a meta material (MM) based SRR type THz-biosensor for the detection of BSA. This sensor consisted of a high resistance Si- substrate, Al- bar resonator and a Al-SRR. The fr of this sensor shifts with reaction time and the RI of the analyte. The usage of nano-meta materials like nano antenna can efficiently enhance the sensitivity of THz-biosensors [56]. In 2018, Banxian Ruan et al., [57] reported a double layer graphene based SPP-THz biochemical sensor that consisted of Ge-prism, two graphene layers, two TPX (Poly methyl pentene) cladding layers and a quartz crystal core layer. This sensor had two modes of operations namely PWG (planar waveguide) and LRSP (long range SPP) modes. LRSP mode is supported by the two parallel layers of graphene and the PWG mode is supported by dielectric layer. The SPP- based THz- graphene biosensor offers high sensitivity and label free detection due to the fact that the angle of resonance of SPP can be modified drastically with a small variation in RI of the sensing medium.

In 2019, A. Samy Saadeldin et al., [58] reported an innovative MM absorption based THz biosensor which was made of two gold layers isolated by a dielectric (teflon) film. Line gaps, line width and thickness of dielectric layer are the key parameters that affects the sensitivity of this sensor. When line width increases, MM-absorption increases and sensitivity decreases.

In 2019, Rui Zhang et al., [59] reported a MM-based micro Fluidic multi channel THz- biosensor for the detection of BSA molecules from small volume solution samples. This sensor was made of bow-tie MM-array, gold metal and a thick quartz crystal wafer. The micro Fluidic multi channels were formed in the middle of bow-tie MM-array. This sensor can also be used for detecting IPA (Iso propyl alcohol). An increase in IPA concentration leads to the decrease of fr shift and increase of absorption dip amplitude. On the other hand, when BSA concentration increases, that also leads to the increase in the fr shift. In 2020, Dongxia Li et al., [60] reported a THz-MM-based biosensor which can be used to detect the thermally induced denaturation temperature of a protein called insulin. This THz-biosensor consisted of a high resistance Si-wafer, a H-shaped metallic architecture enclosed on a U-shaped metallic architecture. The THz waves are allowed to hit on the top of the biosensor. Sensitivity of this biosensor can be computed as

$$S = \frac{\Delta f}{\Delta n} \quad (1)$$

Here, Δf and Δn indicates the shift in fr and RI change. It was observed that the Δf increases with increase in insulin concentration and also denaturation begins from 70⁰ C and finishes at 88⁰ C.

In 2020, Mingjie Tang et al., [61] reported a label free THz-MM-based biosensor which can be used to sense fatty acids such as γ -linoleic acid, α -linoleic acid, linoleic acid and oleic acid. Fatty acids play a crucial role in tissues of living organisms and as ingredients of lipids in membrane- bound enzyme activities, integrity and fluidity. This THz- biosensor consisted of five metallic square array rings, Si-wafer, thick Cr and Au films and SRRs. The fr of this sensor shifts with increase in dielectric constants of the fatty acids in THz regime. Ruijian Cheng et al., [62] reported a Fano resonator type THz- biosensor for the detection of proteins such as AIG and AIG+IgG. In 2021, Nidhi Pandit et al., [63] reported a label free and non intrusive highly sensitive biosensor works based on LSP (localized spoof surface plasmon) resonator with micro Fluidic approach which is capable of detecting aqueous bio samples. This THz- biosensor consists of a LSP ring resonator and a dielectric wafer. The fr of this sensor was found to be increasing with

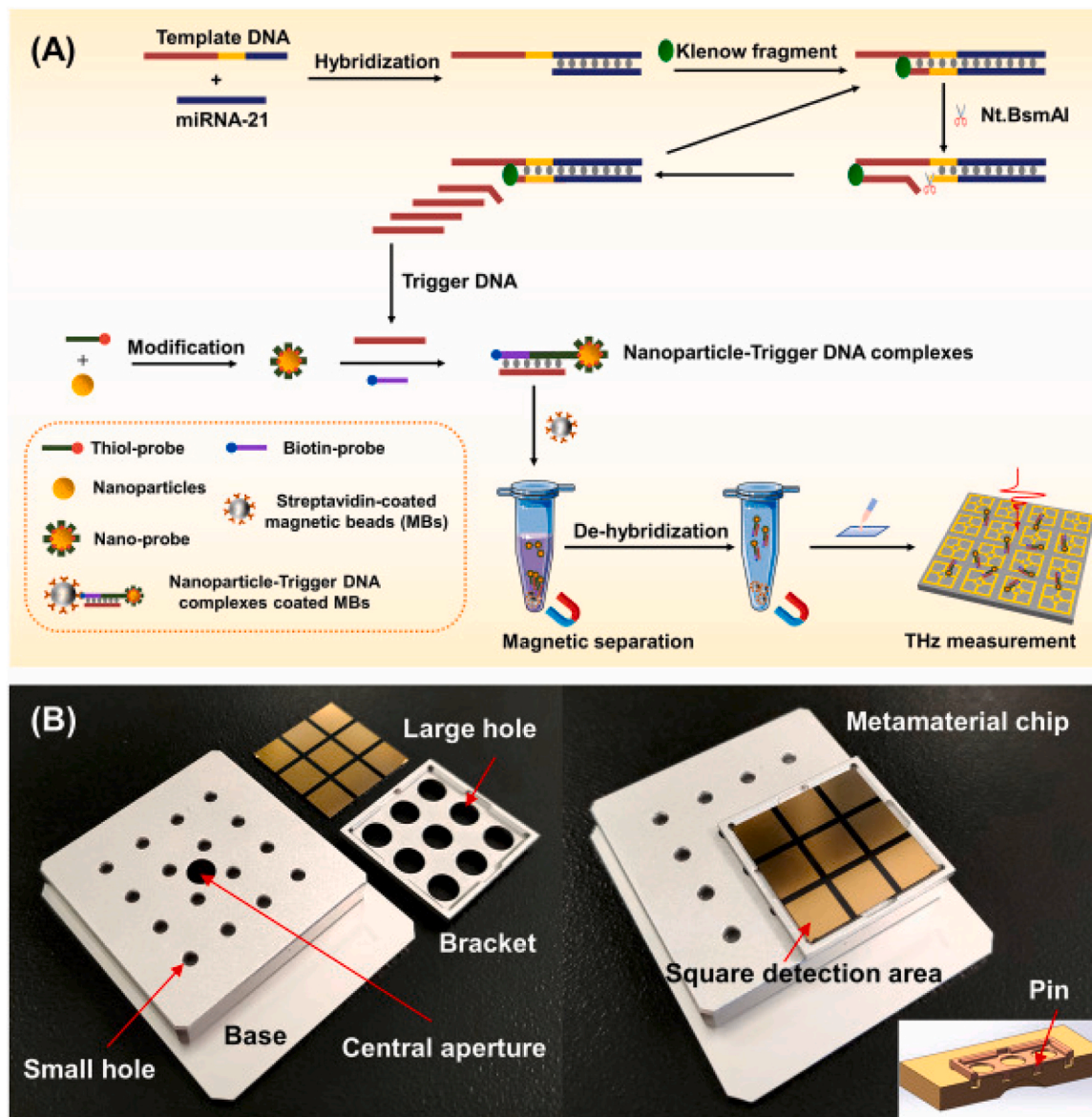


Fig. 3. MM-based THz- biosensor for detecting micro RNAs (Reprinted from [64] with permission from Elsevier).

increase in aqueous bio-sample concentrations. In 2021, Ke Yang et al., [64] reported the development of a MM-based THz- biosensor which can detect micro RNAs. micro RNAs have been considered as a group of biomarkers for diagnosis, prognosis and therapeutic monitoring of diverse cancers this sensor was made of a planar array of SRRs on highly resistive Si-wafer (Fig. 3).

Micro fabrication methods were used to fabricate this sensor. When this sensor is exposed to micro RNA samples, it's frequency shift increases. In the same period, Ruiyun Zhou et al., [65] reported a micro Fluidic label free graphene meta surface based THz-biosensor for the detection of DNA (Fig. 4). This sensor was made of a CASR (complementary asymmetry split ring) graphene meta surface with micro fluidic platform and the CASR meta purpose is formed of a thick gold film and a quartz wafer. Standard photo lithography techniques were used to form CASR graphene arrays on gold films. In 2021, Shuo Liu et al., [66] demonstrated the detection of blood components such as HB, RBCs and WBCs with the help of D- shaped SPR- based THz- biosensor on MoS₂. This sensor had a D-shaped photonic crystal fiber with Zeonex background and a MoS₂ layer on top of the photonic crystal. This THz- bio sensor offered a FoM of 538.98, 865.40, 944.91, 508.90 and 239.45 for

blood components water, plasma, WBCs, HB and RBCs, respectively. In 2022, Jianjun Liu et al., [67] reported a graphene meta material absorber (MMA) based biosensor for detecting an antibiotic called josamycin. The MMA of this sensor had a three layer heterostructure formed by Gold, polyimide and Aluminium layers. Monolayer graphene was deposited on top of gold/polyimide/aluminium heterostructure. The graphene/gold/polyimide/Aluminium heterostructure is exposed to josamycin its reflectivity changes. In the same year, Haiyun Yao et al., [68] reported an innovative flexible THz- meta surface oriented flexible biosensor for the detection of plant protein with a limit of detection of 42.3 pg/mL. Graphene's fermi level shift to the Dirac point when plant protein concentration rises, bonding (covalent) of the structured graphene become stronger and conductivity drops as a result. Therefore, all frequency points shifts and there is a noticeable improvement in the transmission amplitude. Meta surfaces are 2D- materials made of periodic sub wavelength unit geometric cells. As meta surfaces sensitive to variations in their micro environment, external stimuli cause them to respond with a specific resonance that can be utilized in developing biosensors. This flexible sensor consisted of a quartz wafer with spin coated polyamide film on top, two split oval ring resonators and a cut

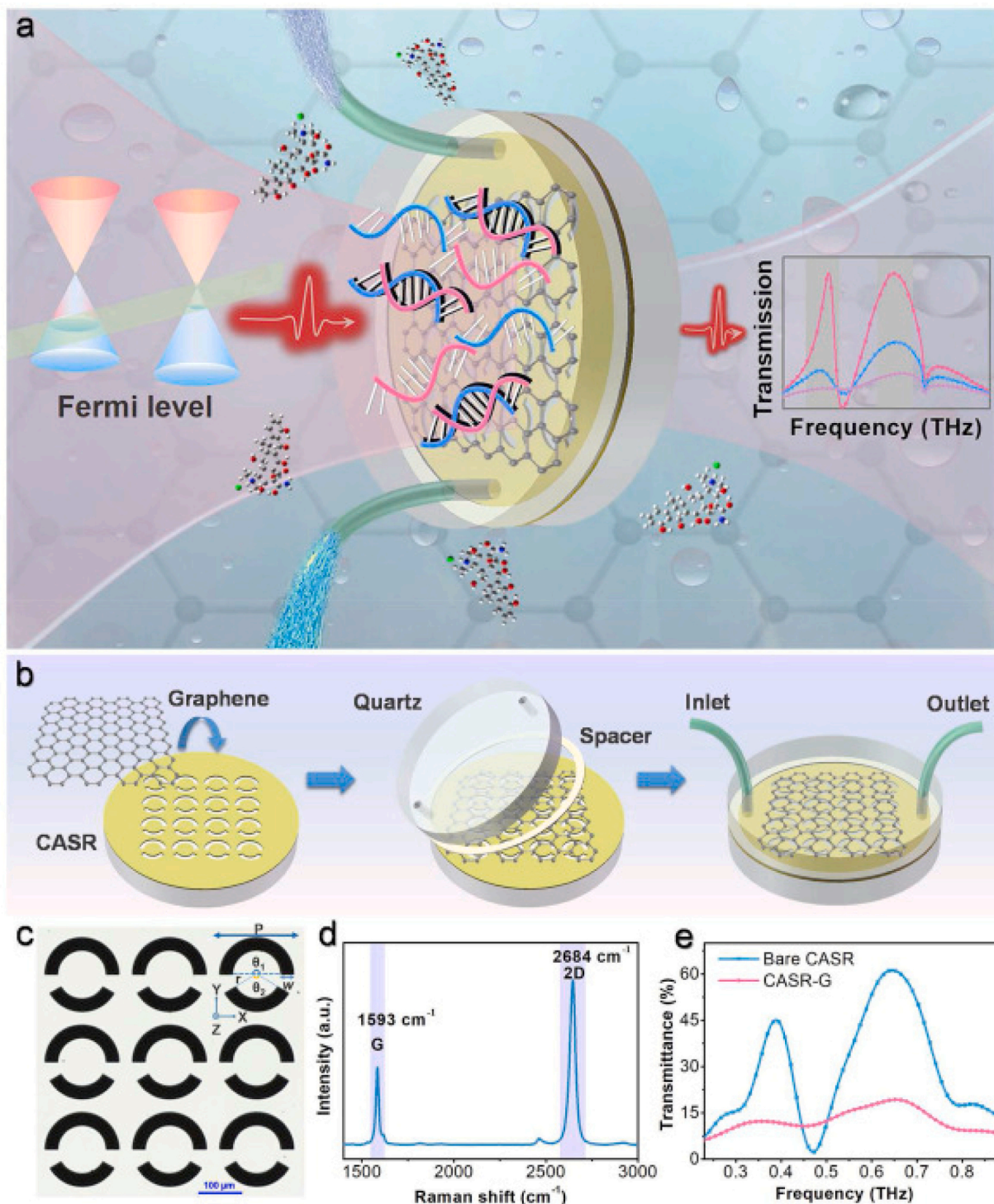


Fig. 4. Micro Fluidic label free graphene meta surface based THz-biosensor for the detection of DNA (Reprinted from [65] with permission from Elsevier).

wire resonator on the polyimide film and a tri-layer graphene.

In 2023, A. N. Nithish et al., [69] reported a hollow core rectangular PCF-THz spectroscopy based biosensor for the detection of hormones such as progesterone and estradiol. In the Healthcare and biomedical field, hormones play a crucial role in the human body. Dangerous disorders including cardiovascular system hyperplasia, osteoporosis, cancer etc are caused by hormone imbalances. The hormones estradiol and progesterone in particular interact with every cell and tissue in female bodies. The changes in female bodies that take place during pregnancy

are determined by the interaction between these two hormones. Hence, it is essential to identify the levels of estradiol and progesterone hormones during pregnancy time. Weijian Cai et al., [70] fabricated a double split (DS)- hexagonal ring (HR)- meta material based highly sensitive THz-biosensor for the detection of bovine serum protein (BSA) and glucose that shown a sensitivity/limit of detection of 36.44 GHz/(mg/mL)/0.16 mg/mL, respectively. Thickness of the analyte, dielectric permittivity and sample concentration are the factors that affects the shifting of frequency in MM-based THz- biosensors. The

Table 1

The sensing structures/mechanisms and performance of THz-biosensors (PIT: Plasmon-Induced Transparency).

Ref.	Sensing Structure/ Mechanism	Sensitivity	Q- Factor	FoM
[54]	Split Ring Resonator	$3 \times 104 \text{ nm/RIU}$	41	-
[53]	Hybrid plasmonic ring resonator	$46 \mu\text{m/RIU}$	550	82
[57]	Surface plasmon polariton	292/RIU	-	1.79
[58]	Meta Material Absorber	300 GHz/RIU	22.05	2.94
[59]	Meta Material Absorber	552 GHz/RIU	-	1.58
[51]	Graphene	4.2 THz/RIU	-	12.5
[51]	Subwavelength	7.6 THz/RIU	-	12.5
[51]	Plasmonic grating	10 THz/RIU	-	24.5
[52]	Double ring resonator	3260 nm/RIU	1432	62.81
[50]	Graphene-based Bragg reflector	517.90/RIU	-	222.9/RIU
[60]	Meta Material Split ring resonator	85 GHz/RIU	-	-
[62]	Fano Resonator	160 GHz/RIU	5.67	0.91
[75]	Meta Material Absorber	1.6 THz/RIU	225	80
[76]	Fano Resonator	96480 nm/RIU	225	195
[77]	Fan-shaped resonator	1550 GHz/RIU	6.66	2.87
[78]	Dual split ring resonator	930.4 GHz/RIU	23.7	12.53
[79]	Fano Resonator	16042 nm/RIU	39857	533
[63]	Spoof Surface Plasmon	13.5 MHz/mgml-1	192	-
[66]	Surface plasmon resonance	715.59 $\mu\text{m/RIU}$	-	538.98
[80]	Spoof plasmon resonance	$2.14 \times 10 - 3 \text{ THz/er}$	539	-
[81]	Photonic crystal resonator	45 MHz/ μm	5063	3.1
[82]	Meta Material Absorber	56.67 THz/RIU	-	358.7
[83]	Meta Material Absorber	1.762 THz/RIU	444	392
[84]	Plasmonic meta surface	265.06 GHz/RIU	64	9.1
[85]	PIT-Graphene Metamaterial	1.1 THz/RIU	-	3.18
[86]	Toroidal Dipole Metasurface	775.7 GHz/RIU	1016	284
[87]	Spoof Surface Plasmon	517.9 GHz/RIU	262	-
[70]	Double-split hexagonal ring	430 GHz/RIU	-	-
[71]	EIT-Meta surface	13.5 MHz/mg/mL	-	-
[73]	EIT-Meta surface	333.33 GHz/RIU	7.23	3.11
[88]	Meta Material Absorber	196.5 GHz/RIU	165	19.65
[89]	Dual plasmon-induced transparency	1.3 THz/RIU	37	6.259
[74]	EIT-Meta surface	175 GHz/RIU	17.62	4.52
[10]	Metal meta surface	191.3 GHz/RIU	16.9	6.31
[9]	Fano Resonator	420 GHz/RIU	58	209
[69]	PCF-Based THz-Spectroscopy	99.87%	-	-
[69]	PCF-Based THz-Spectroscopy	99.88%	-	-

DS-HR-MM sensors consisted of quartz wafer, with gold coating. Weijian Cai et al., reported that the changes in glucose & BSA concentration leads to the shifting of frequency of the sensor.

In the same year, Lanju Liang et al., [71], Jianyuan Qin et al., [72], S. K. Patel et al., [73] and R. Jadeja et al., [74] reported THz-biosensors for the detection of aspartic acid, saccharides, amino acids and peptides, respectively. Lanju Liang et al., [71], demonstrated a graphene/GaN flexible meta material hetro structure-based THz-biosensor with a detection limit of 10.6 fg/mL for the detection of aspartic acid. This aspartic acid sensor was made by polyimide coated Z-cut quartz wafer, two split semi-circle resonator (SCR) and a cut wire resonator. In the sensor, THz-waves can be applied to the EIT- meta surface in the Z-axis direction. The THz-waves directly excites the SCR which leads to a dipole resonance at 1.47 THz. The THz waves also excites the cut wire resonator directly and produces a dipole resonance at 0.92 THz. Both the resonators functions as bright modes only. In cut wire resonator, the electric field will be distributed at the ends of the rectangular bars. On the other hand, in SCR resonator, electric field will be distributed mainly in both semi circles. This leads to the generation of transparent Windows

which are highly sensitive to external stimuli. S. K. Patel et al., [73] reported a EIT- meta surface based THz-biosensor which can be used for detecting amino acids such as glycine, glutamine, asparagine, alanine, glutamic acid, α - amino butyric acid, lysine, α -amino valeric acid, arginine, Leucine, Histidine, isoleucine, tryptophan, serine, phenylalanine, threonine, methionine and proline. The RI of these amino acids lies between 1.5 to 1.8. Different amino acids exhibit different sensitivities and resonant frequency shifts. The sensing structures/mechanisms and performance of THz-biosensors are listed in Table 1.

THz waves are highly sensitive to weak intermolecular-interactions, including van der Waals forces and hydrogen bonds. Therefore, THz spectroscopy is one of the most promising biological sensing techniques because most chemical & biological molecules (DNA, RNA, and protein) have fingerprint vibrational frequencies that fall within this spectral range. The RI of some biomacromolecules, such proteins or RNA or DNA, and small biomolecules, like fungal bacteria, with thicknesses up to 50 μm , fall between 1 and 2. As a result, biomolecules may be detected and distinguished with extreme sensitivity. Label-free THz-sensing detects target-molecules directly through RI changes caused by molecular-interactions with surface plasmons, which are related to molecular weights or molecular-structures or molecular-dipole-polarization-reactions or surface density or sample concentration. As a consequence of this, the detection signal is not diminished with sample volume.

Label-free THz-biosensors have tremendous potential in the health-care/ biomedical applications in the coming years because these sensors are cheap and easily operable. Various sensing structures have been reported to develop label-free THz-biosensors such as SRRs, hybrid-plasmonic-ring resonators, MMAs, sub-wavelength structures, plasmonic-grating, double-ring resonators, graphene-based Bragg-reflectors, metamaterials, PCRs, plasmonic-meta surfaces and so on. THz-SPPs have lower propagation loss than visible range SPPs, but have poor confinement on metal surfaces. Therefore, THz-SPPs are unsuitable for compact integration. This issue can be effectively tackled by replacing metal surfaces with graphene strips. Graphene is widely used in the sensing structures due to recent advances in 2D-materials, which have boosted the sensing capabilities. When it comes to detecting viral & bacterial diseases, THz-biosensors are incredibly fast and precise.

3. Terahertz chemical sensors

THz-sensors have been reported for sensing chemicals such as ethanol [90], [91] benzene [92], [93], water [94], [95] hydrogen sulfide [96], hydrogen cyanide [96], ethylene glycol [96], dichlorofluoro methane [96] toluene [96], food additives such as $\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5$ (Aspartame) & $\text{C}_8\text{H}_8\text{O}_3$ (Vanillin) [97], ammonia & urea [98], progesterone & estradiol [69], organic compounds such as ethyl acetate (EA), iso propyl alcohol (IPA) and ethyl benzene (EB) [99], pesticide, imidacloprid [100] and chiral compounds such as L and D tartaric acids [101] and so on. In 2018, B. K. Patel et al., [90] reported a QPCF (quasi photonic crystal fiber) based THz- spectroscopic chemical sensor for the detection of water, benzene and ethanol. This QPCF-based THz-chemical sensor consisted of a porous Core air holes (circular shaped) and a quasi shape cladding. The confinement loss (L_c) occurs from the cladding as well as core porosity can be estimated as

$$L_c = \frac{4\pi f}{C} I_m n_{eff} \quad (2)$$

Here, I_m , f and C indicates the propagation constant (imaginary part), applied frequency and velocity of light, respectively. The operating frequency effects on relative sensitivity of QPCF-based THz- chemical sensor is depicted in Fig. 5(a). For water, benzene and ethanol, the relative sensitivity of the sensor initially rises from 0.8 THz to 1.2 THz and after that it became constant and independent of applied frequency.

Confinement loss (Fig. 5(b)) for QPCF-based sensor with water, benzene and ethanol decreases from 0.8 THz to 1.3 THz and there after

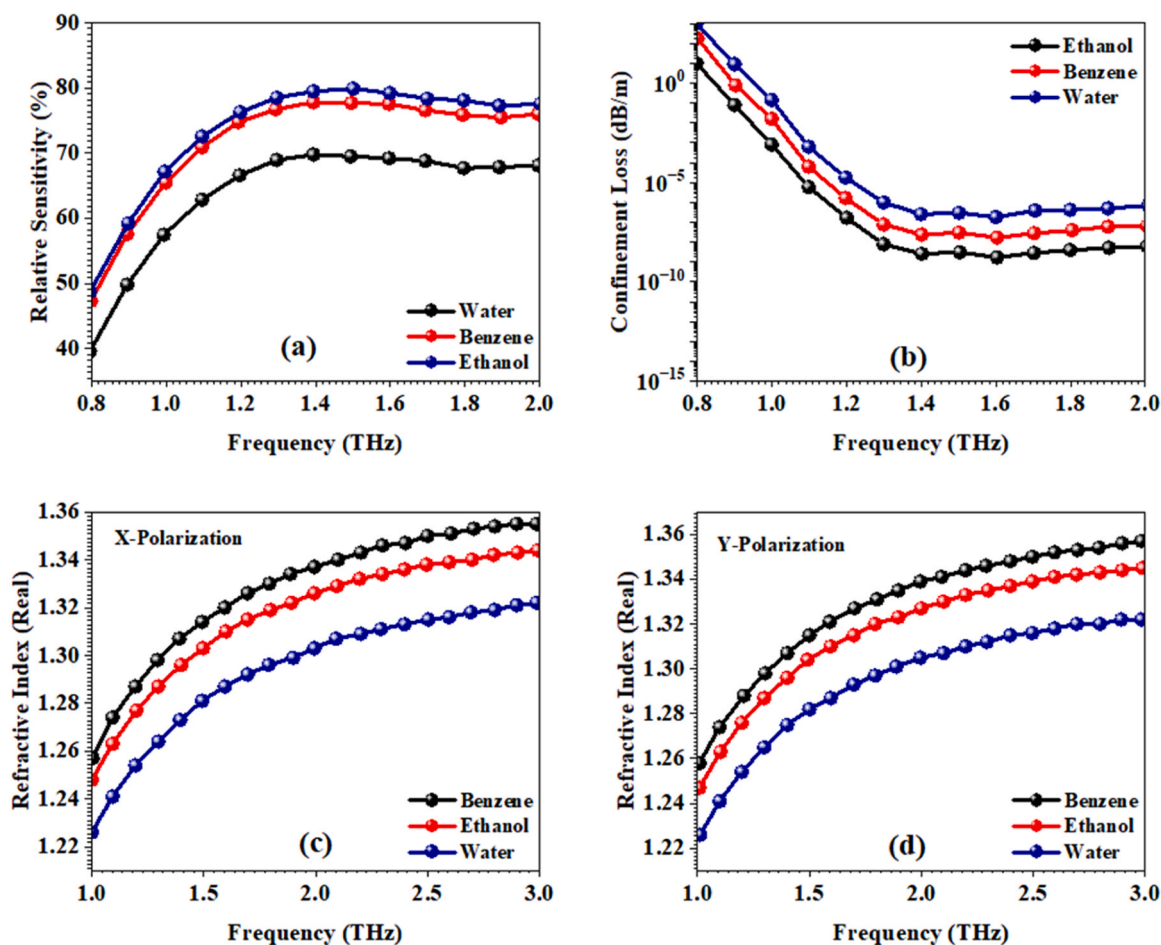


Fig. 5. (a) & (b) QPCF-based THz- chemical sensor- operating frequency effects on relative sensitivity & confinement loss [90] (c) & (d) The changes in effective RI against applied frequency of benzene, water and ethanol solutions under x & y polarizations [95].

it was independent of applied frequency. In 2020, Fuyu Li et al., [92] demonstrated a THz-sensor Which can detect ethanol solutions. This sensor featured a double ring resonator that offers EIR (electromagnetic induced reflection) Fano resonance and the sensitivity of this sensor can be modified by adjusting gap width. In 2021, Ramachandran et al., [95] reported an innovative DSW (dual steering wheel)/PCF-based THz-chemical sensor for the detection of ethanol, benzene and water featuring non circular and rectangular asymmetric air holes in the core. These non circular air holes enhances the sensitivity of PCF-based THz-sensors. The structure of a PCF-based THz- chemical sensor which can be used for detecting ethanol, benzene and water are depicted in Fig. 6. A light source such as laser can be used to apply THz-waves. The changes in effective RI against applied frequency of benzene, water and ethanol solutions under x & y polarization are depicted in Fig. 5(c) & (d). The RI of benzene, ethanol and water rises exponentially with applied frequency from 1.1 THz to 3 THz.

In 2021, M. H. K. Anik et al., [96] reported a HCTW (hollow core THz- waveguide)-PCF-based THz-chemical sensor which can be used to detect chemicals such as ethylene glycol, dichloro fluoro methane, hydrogen cyanide, hydrogen sulfide, and toluene. Usually, polymers such as TOPAS, teflon, Zeonex, Arton, Apel and PMMA are used as background materials of the fibers. Good optical transmission, low moisture uptake, low absorption loss and high chemical resistance are the required properties of these polymers. Among the above mentioned background polymer materials, TOPAS is the best one. The relative sensitivity of this THz- chemical sensor was founded to be rising with increase in applied frequency from 1.2 THz to 2.2 THz regime.

Food safety if an essential aspect for everyday life and national

productivity. Food safety has an impact on public health. The modern Food Industry cannot function without food additives. However, there is mounting evidence that too many food additives are harmful to the health. As a result, more emphasis should be given to the identification of harmful food additives. Hyper spectral imaging, liquid chromatography and near infrared (NIR) spectroscopy are the most widely used methods for sensing food additives. However, these sensing methods have the limitations such as long detection time and complicated sample pre treatments. In 2022, Jinhui Zhang et al., [97] reported a MMA-based THz- chemical sensor which can effectively detect food additives such as aspartame (artificial sweetener) and vanillin (for fragrance). Aspartame usage has serious cellular toxicity particularly to the kidneys and liver, as well as impairments to gluconeogenesis and other processes. Similarly, vanillin overuse can cause kidney injury, headaches and insomnia. The THz- chemical sensor reported by Jinhui Zhang et al., [97] consist of a ring resonator with polyimide wafer, a dielectric with permittivity 2.98 and aluminium rings. When exposed to food additives, the transmission peak of the sensor changes with applied THz-waves. VOCs (Volatile organic compounds) are the by products produced by petroleum, chemical, paint and food industries. These VOCs are hazardous to environmental safety and human health. VOCs can adversely affect respiratory, immune, cardiovascular, reproductive, nervous and circulatory systems of human body. VOCs can also affects organs such as spleen, stomach, liver and Kidneys. In 2023, Wenfeng et al., [99] developed a THz- chemical sensor works based on EIT- meta surface which can effectively identify the VOCs. The sensing structures/mechanisms and performance of THz-chemical sensors are listed in Table 2.

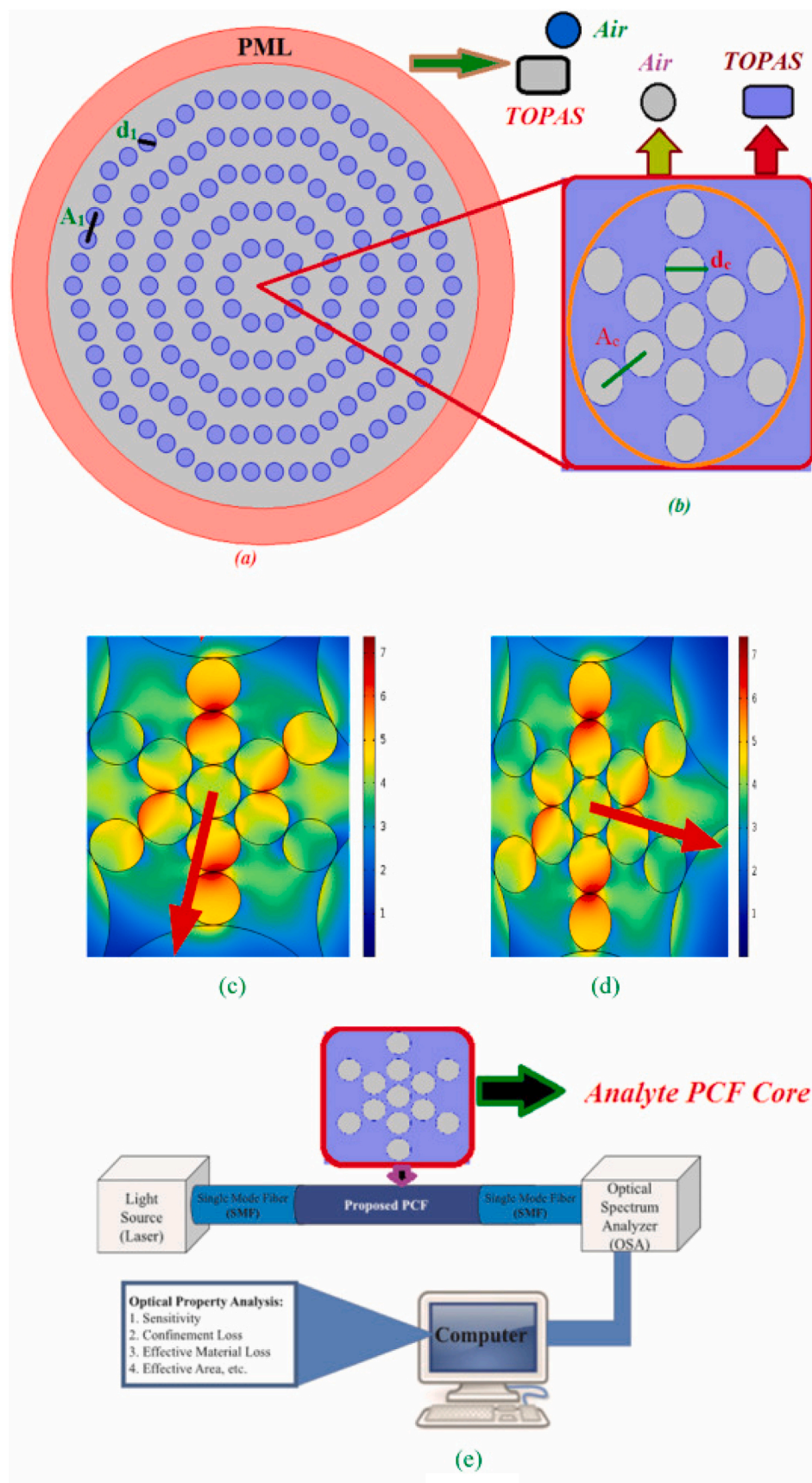


Fig. 6. The structure of a PCF-based THz- chemical sensor which can be used for detecting ethanol, benzene and water (Reprinted from [91] with permission from Elsevier).

Table 2

The sensing structures/mechanisms and performance of THz-chemical sensors.

Ref.	Sensing Structure/ Mechanism	Chemical to be detected	Sensitivity (%)	Confinement Loss (dB/m)
[90]	QPCF-based THz-spectroscopy	Ethanol	78.8	7.15×10^2
[90]	QPCF-based THz-spectroscopy	Benzene	77.8	1.80×10^2
[90]	QPCF-based THz-spectroscopy	Water	69.7	7.51×10^2
[93]	PCF-based THz-spectroscopy	Ethanol	97.9	6.95×10^{-14} /cm
[93]	PCF-based THz-spectroscopy	Benzene	98.2	3.48×10^{-14} /cm
[93]	PCF-based THz-spectroscopy	Water	97.6	6.22×10^{-13} /cm
[92]	Meta Material Absorber	Ethanol	112 GHz/RIU	-
[94]	PCF-based THz-spectroscopy	Ethanol	68.59	10^{-11}
[94]	PCF-based THz-spectroscopy	Benzene	70	10^{-11}
[94]	PCF-based THz-spectroscopy	Water	67.62	10^{-11}
[95]	DSW-micro-structured PCF	Ethanol	95.4	-
[95]	DSW-micro-structured PCF	Benzene	95.6	-
[95]	DSW-micro-structured PCF	Water	94.9	-
[91]	Hexahedron Core PCF	Ethanol	92.55	5.80×10^{-8}
[91]	Hexahedron Core PCF	Benzene	94.65	6.01×10^{-8}
[91]	Hexahedron Core PCF	Water	90.30	5.75×10^{-8}
[96]	Rectangular shaped HCTW-PCF	Toluene	99.73	10^{-11}
[96]	Rectangular shaped HCTW-PCF	Ethylene glycol	98.85	10^{-11}
[96]	Rectangular shaped HCTW-PCF	Hydrogen Sulphide	97.22	3.2×10^{-11}
[96]	Rectangular shaped HCTW-PCF	Dichloro Fluoro Methane	93.02	10^{-11}
[96]	Rectangular shaped HCTW-PCF	Hydrogen Cyanide	89.56	10^{-11}
[97]	Meta Material Absorber	Aspartame	30.76 dB/(ng/mm ²)	-
[97]	Meta Material Absorber	Vanillin	63.91 dB/(ng/mm ²)	-
[99]	EIT-Meta Surface	Iso Propyl Alcohol	5.45 µg	-
[99]	EIT-Meta Surface	Ethyl Benzene	4.35 µg	-
[99]	EIT-Meta Surface	Ethyl Acetate	13.46 µg	-
[100]	Meta Material Absorber	Imidacloprid pesticide	220.7 GHz/RIU	-
[101]	SWNT- Meta Material Absorber	Chiral molecules	-	-

THz-chemical sensors commonly use PCF, MMAs and EIT-metasurfaces as sensing structures. Among these sensing structures, PCF-waveguide based THz-chemical sensors have received incredible attention in recent years due to their high bit-rate transmission of data high resolution, excellent design flexibility, compact size, low limit of

detection and higher sensitivity. Further more, PCF-based THz-chemical sensors can tolerate hazardous chemicals, high voltages and high temperatures. For chemical sensing, hollow core PCF preferred over elliptical, octagonal, hexagonal, solid core and porous core PCF because it provides a larger volume of analyte inside its core than the other types of cores. Additionally, the hollow core guided modes are highly confined inside the core region, thus minimizing the impact of the background material on the PCF's waveguiding characteristics. PCFs that operate based on the SRR-principle exhibits changes in the analytes RI as a predominant sensing response.

IV. Terahertz sensors for disease detection

THz-technologies have proven to be extremely useful to the Healthcare and medical fields since a great deal of biomolecules have distinct spectral fingerprints in the THz-regime. This attribute along with the radiation's non-ionizing nature made THz-based sensing a key technology for a extensive variety of biomedical/ healthcare applications, including non destructive thin film biomolecular sensing for disease screening and diagnosis. THz-sensors have been reported for the detection of various diseases such as skin cancer [102], oral cancer [50], colon cancer [103], breast tumor [104–111], cervical cancer [112], blood cancer [113], lung cancer [114], heart failure [115], a tropical disease called African trypanosomiasis [116] and liver cancer [117]. HAT (human African trypanosomiasis) is a deadly tropical illness triggered by a single cell parasite in blood. In 2017, Mario Mueh et al., [116] reported a THz-sensor for the efficient detection of this tropical disease. The THz-sensor used for detecting HAT disease consist of aDSR (asymmetric double split ring resonator). The sample needed to be placed in the path of THz-beam. This sensor operation had two modes-one, trapped current mode (at 335 GHz) and a electron dipole mode [at 495 GHz]. Periodicity, ring radius and width was optimized to fix the dipole and trapped modes between 250–600 GHz. The trapped mode provides high sensitivity compared with dipole mode.

In 2019, A. Keshavarz et al., [102] demonstrated a THz-sensor for the detection of skin cancer. Skin cancer is one of the most deadly diseases that can lead to death. Skin cancer can be easily treated if identified early, but if detected late, not only is treatment more challenging, but it can also spreads to other body parts quickly. Many scientists are currently working on early diagnosis of skin cancer. OCT (optical coherent tomography), spectral analysis, thermal imaging and Raman micro spectroscopy are the most commonly used to techniques to the detection of skin cancer. A. Keshavarz et al., [102] developed a water based THz-MM-based sensor for the detection of skin cancer. The MM-structure of this sensor consist of a pair of water rods and a InSb square film. When the THz-waves are exposed to the sensor, the reflection spectrum of the sample changes.

In 2021, Z. Vafapour et al., [103], A. Hlali et al., [104] and Dongxia Li et al., [112] reported the detection of colon cancer, breast tumor and cervical cancer, respectively using THz-sensors. Z. Vafapour et al., [103] reported a water based THz-sensor featuring a water filled InSb ring resonator, a glass substate and InSb meta material and MgF2 buffer layer. This sensor works based on SPP. When THz-waves are exposed to sample solution, the spectral response of the sensor changes. A. Hlali et al., [104] reported a WCIP (wave concept iterative process) technique based to tunable graphene THz-sensor for the detection of breast tumor. This sensor Was made of a patch antenna, graphene layer metal layer and a dielectric layer. Xin Yan et al., [118] reported a EIT-like MM-based THz-sensor for the detection of oral cancer. This sensor featured double SRR and polyimide dielectric layer (Fig. 7). When THz-waves are exposed to the sensor, these waves will penetrate the meta materials from the dielectric layer to the metallic layer on which a large number of cells have been attached. The cancer cell concentration leads to the shifting of resonance frequency and transmission spectral characteristics. A. Veeraselvam et al., [113] reported a THz sensor for the detection of blood cancer. This SPR-based sensor was made of meandered stubs

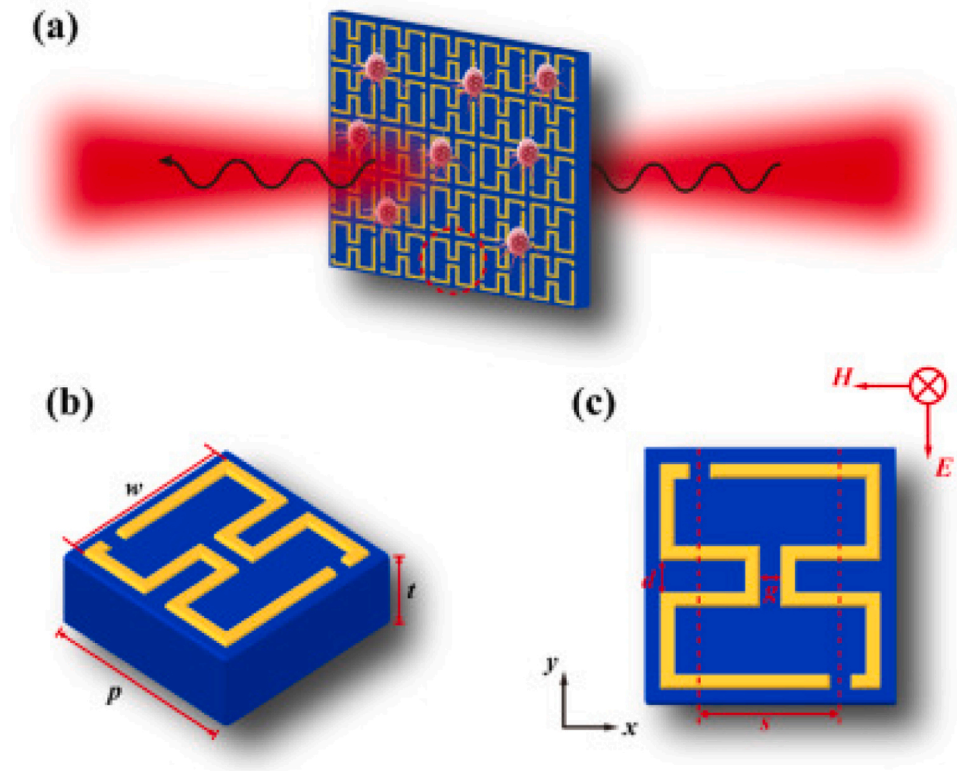


Fig. 7. THz- sensor for the detection of oral cancer (Reprinted from [118] with permission from Elsevier).

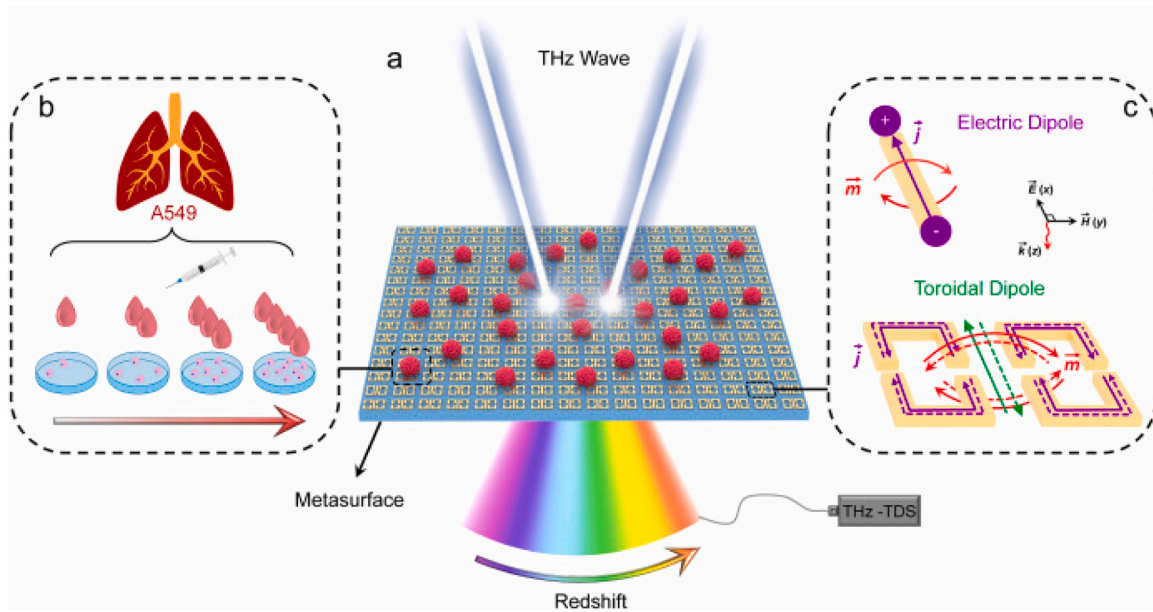


Fig. 8. THz- sensor for the detection of lung cancer (Reprinted from [119] with permission from Elsevier).

embedded in a square shaped loop. This sensor was made on silicon wafer with bottom and top Cu- conducting layer. In between Cu-layers a polyimide layer is enclosed. When exposed to THz- waves, its transmission, reflection and absorption characteristics of this sensor will be changed and that leads to the shifting of resonant frequency of the sensor. In the same year, Chiban Zhang et al., [119] reported a meta surface based THz- sensor for the detection of lung cancer (Fig. 8). This

sensor featured a couple of mirrored Au-SRRs and a glass substrate. A cut wire is inserted between these two resonators. When exposed to THz-waves, this THz- sensor experience a change in transmission spectrum characteristics.

In 2023, Wentao Liu et al., [114] reported a THz- sensor based on meta surface resonance for the detection of SCCA (Squamous cell carcinoma antigen) levels which is a strong indicator of lung cancer. This

Table 3

The sensing structures/mechanisms and performance of THz-sensors for the detection of diseases.

Ref.	Sensing Structure/Mechanism	Disease to be detected	Sensitivity
[116]	Asymmetric double SRR	HAT	6.86%
[102]	MMA	Skin cancer	117 $\mu\text{m}/\text{RIU}$
[103]	MMA	Colon cancer	-
[104]	WCIP	Breast cancer	8.21 THz/RIU
[112]	MMA	Cervical cancer	74 GHz/RIU
[118]	EIT-Meta material induced resonance	Oral cancer	900 kHz/RIU
[105]	MMA	Breast cancer	1500 GHz/RIU
[105]	MMA	Cervical cancer	1541.66 GHz/RIU
[120]	Meta surface resonance	Cancer	155 GHz/RIU
[106]	MMA	Cancer	1447 GHz/RIU
[107]	THz-waveguide Iris probe	Breast cancer	0.2 mm
[113]	SPR	Blood cancer	514.28 GHz/RIU
[113]	SPR	Skin cancer	476.35 GHz/RIU
[113]	SPR	Breast cancer	476.91 GHz/RIU
[119]	Meta surface resonance	Lung cancer	403.7 GHz/RIU
[108]	Meta surface resonance	Breast cancer	108 GHz/RIU
[109]	THz frequency-comb-like	Cancer	105.004 GHz/RIU
[114]	Meta surface resonance	Lung cancer	103 GHz/RIU
[117]	MMA	Liver cancer	68 GHz/RIU
[110]	Meta material cavity	Breast cancer	97 GHz/RIU
[110]	Meta material cavity	Skin cancer	105 GHz/RIU
[111]	Meta material refractometer	Cancer	495 GHz/RIU
[115]	THz-spectroscopy	Heart failure	98.37%

lung cancer THz- sensor featured a double SRR. Double the three layers of this sensor were made by using Au, SiO₂ and Si-materials. When THz-waves are applied to the meta surface of the sensor, the transmission spectral characteristics changes. Duan Xie et al., [117] reported an innovative THz- sensor for detecting liver cancer. This sensor identifies HCC (hepatocellular carcinoma) with the antibody against AFP (α -fetoprotein). The sensor featured SRRs array on the SiO₂ surface and a Si-wafer. When a X- polarized THz- waves are applied to the sensor, to resonant frequencies will be obtained in the transmission spectra. This sensor offered good linearity, high specificity, sensitivity and low LOD (limit of detection) which makes them promising for future clinical applications. Ping Yu et al., [115] reported a THz- sensor for the detection of heart failure which works based on the THz- spectroscopy. This sensor detects myocardial amyloidosis which is an indicator of heart failure. The extracellular area of the heart is primary interstitial protein is accumulated and this condition is known as myocardial amyloidosis. It could cause a organ disfunction right away and ultimately result in heart failure. Radio nuclide cardiac magnetic resonance and histo pathology are the techniques commonly used to detect micro cardiac amyloidosis. Large time consumption and dangerous to the patients are the major disadvantages of these methods. Ping Yu et al., [115] used TPI (THz- pulse imaging) technology for identifying the microcardial amyloidosis. The sensing structures/mechanisms and performance of THz-sensors for the detection of diseases are listed in Table 3.

THz-radiation is ideal for detecting various diseases such as cancers, heart failures and HAT due to high resolution its non ionizing properties. THz-sensors utilizes sensing structures/mechanisms such as asymmetric double SRR, MMAs, WCIP, EIT-meta material induced resonance, meta surface resonance, THz-wave guide Iris-probe, THz-frequency comb-like

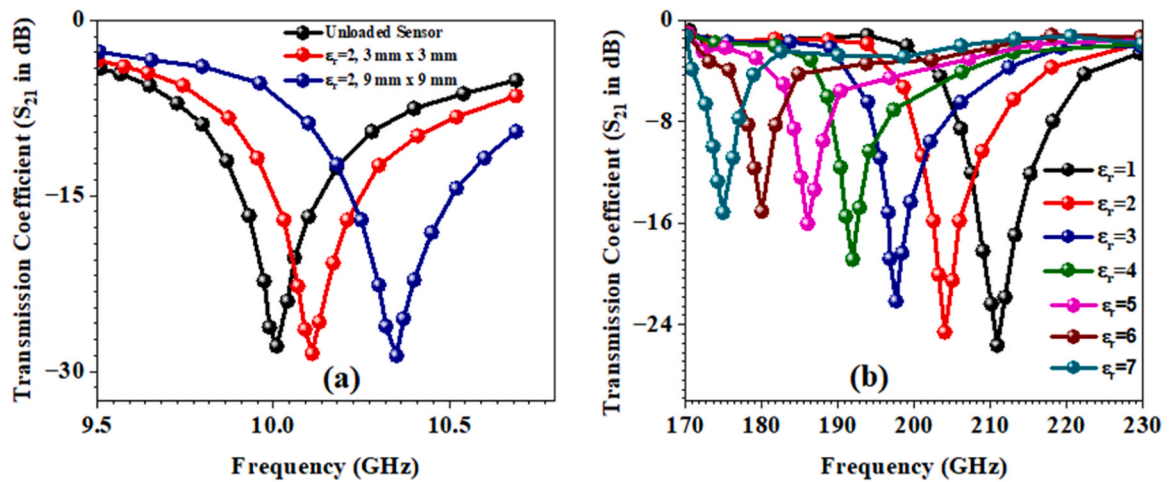


Fig. 9. The transmission coefficient (S_{21} in dB) of the THz- dielectric sensor with loaded and unloaded conditions [121].

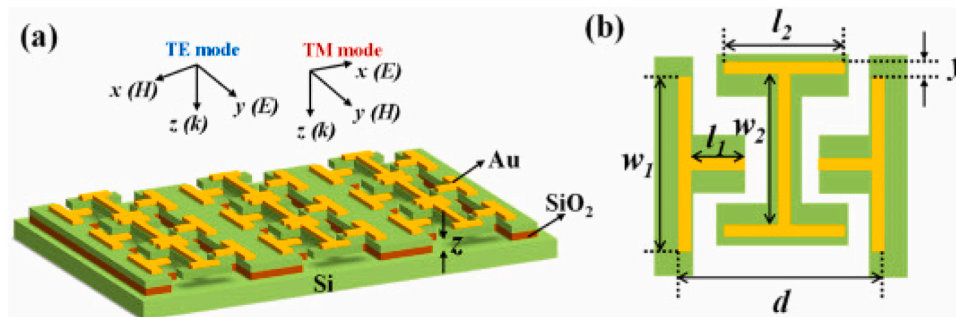


Fig. 10. THz-RI sensor featuring an electric SRR with T & I-shaped resonators on SOI-wafer (Reprinted from [136] with permission from Elsevier).

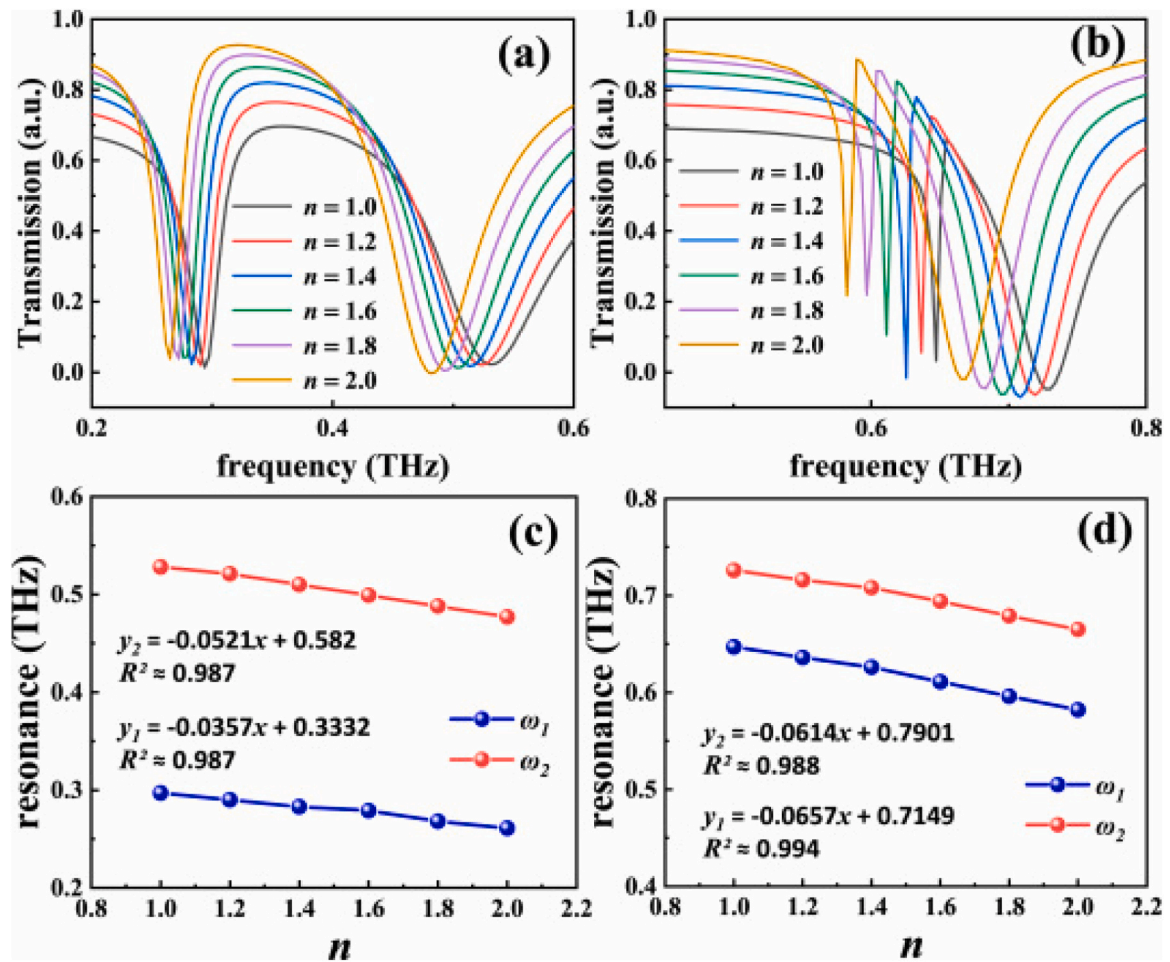


Fig. 11. The transmission spectra of THz-RI sensor featuring an electric SRR with T & I-shaped resonators on SOI-wafer (Reprinted from [136] with permission from Elsevier).

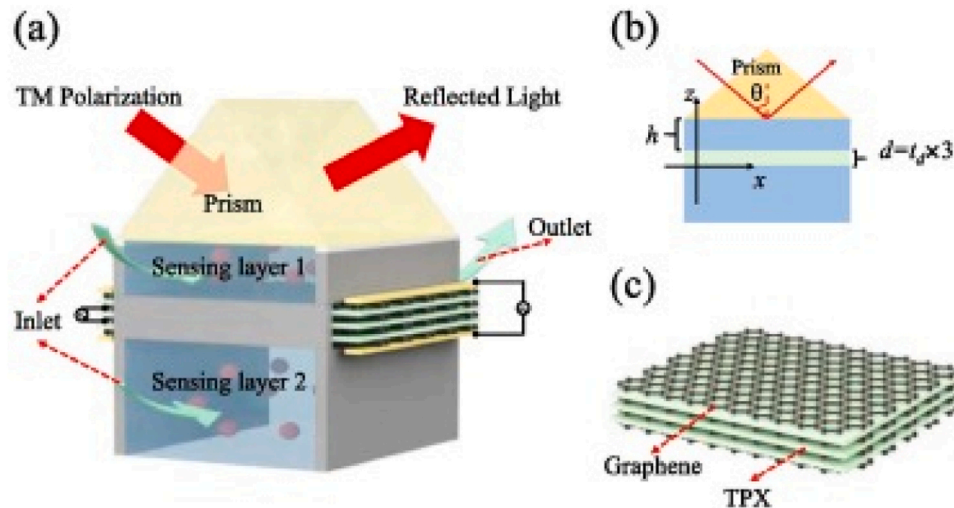


Fig. 12. Dielectric/graphene composite structured meta material based THz- sensor for the sensing of RI of analytes (Reprinted from [140] with permission from Elsevier).

and meta material refractometers for detecting diseases. The combination of THz- waves and meta surfaces is an efficient way to improve the disease detection sensitivity.

5. Terahertz dielectric sensors

Surface bound EM (electromagnetic) waves called SP (surface plasmons) travel along the dielectric- metal interface and exponentially decay in the direction normal to the interface. In 2020, S. P. Singh et al.,

Table 4

The sensing structures/mechanisms and performance of THz-sensors for the sensing of dielectrics & RI of analytes.

Ref.	Sensing Structure/ Mechanism	Mode of operation	Sensitivity	Q-Factor	FoM
[121]	Spoof Surface Plasmon	Transmission	2.84%	165	-
[141]	Graphene meta surface	Reflection	1.5 THz/ Permittivity unit	-	-
[122]	Spoof Surface Plasmon	Transmission	85.75 ⁰	-	-
[123]	MMA	Reflection	2.2 THz/RIU	927.5	550
[124]	Si-Photonic column array	Transmission	261 GHz/RIU	-	-
[127]	Graphene SPR	Reflection	49.5 ⁰ /RIU	-	8.76
[125]	QBIC-Ring chain Fano resonator	Transmission	420 GHz/RIU	503	8.12
[128]	Plasmonic waveguide	Transmission	0.11 THz/RIU	-	16.75
[129]	Insulator/graphene SPR	Reflection	51 ⁰ /RIU	-	12.75
[130]	MMA	Transmission	22.72 THz/ RIU	-	13.84
[131]	Graphene chiral meta surface	Transmission	7000- 10000 nm/ RIU	-	-
[132]	Square SRR	Reflection	126 GHz/RIU	44.17	10.5
[142]	MMA	Reflection	52.5 GHz/RIU	-	-
[133]	Graphene photonic crystal	Reflection	0.4203 THz/ RIU	280	100
[139]	Dielectric meta surface	Transmission	105 GHz/RIU	412	170.6
[134]	InSb-Plasmonic ring resonator	Transmission	0.509 THz/ RIU	11.62	6.889
[143]	MMA	Transmission	292 GHz/RIU	13.05	172
[143]	MMA	Transmission	780 GHz/RIU	12.82	8
[135]	EIT-like meta material	Transmission	330 GHz/RIU	24	3.54
[144]	MMA	Transmission	100 GHz/RIU	-	7-10
[145]	EIT-like meta material	Transmission	276.6 GHz/ RIU	-	7.782
[146]	Graphene photonic crystal	Reflection	972.5 GHz/ RIU	29410	5587
[147]	EIT-like meta material	Transmission	185 GHz/RIU	-	-
[39]	Distributed Bragg Reflector	Reflection	3 THz/RIU	-	776.17
[136]	SRR	Transmission	65.7 GHz/RIU	-	-
[140]	Graphene/dielectric meta material	Reflection	20.5 THz/RIU	-	27
[148]	MIM – Meta surface	Reflection	572 GHz/RIU	22.8	6.81
[137]	E-Shaped meta surface	Transmission	0.7 THz/RIU	3362	1178
[138]	PIT-based graphene square ring	Transmission	0.7 THz/RIU	-	2.4

[121] reported a SSP (Spoof Surface Plasmon) oriented THz- sensor that can be used to for sensing dielectrics. This sensor consisted of a CPW (coplanar wave guide) line loaded with dual T- shaped metallic periodic array structures. The materials used were Cu and RT duroid 6010LM wafer. The transmission coefficient (S_{21} in dB) of the THz- dielectric sensor with loaded and unloaded conditions are depicted in Fig. 9.

The ϵ_r of the samples can be computed as [121].

$$\epsilon_r = 82.2156 - 12.7615f_r + 0.47329f_r^2 \quad (3)$$

In 2022, Ankita Kumari et al., [122] reported a THz-dielectric sensor that works based on SPP. This differential phase sensors outperforms traditional resonant THz- sensors. In addition to surface plasmon elements, MMAs may also be used to measure the dielectric constant of

samples at THz frequencies. In 2023, Patri Upender et al., [123] developed a MMA- based THz- dielectric sensor that consisted of a Au-layer at the bottom, SiO₂- substrate, a graphene ring and a Si- based dielectric resonator. This dielectric sensor can be used for biosensing and chemical sensing applications such as detection of glucose in water, dengue and malaria etc. The fields of biomedicine & chemistry have recently shown a great deal of interest in THz technology because of its special benefits, which include nonionizing, non-destructiveness and strong penetration. Nevertheless, the sensitivity of earlier metallic or dielectric metasurface THz-sensors is typically low. The detection capabilities THz-dielectric sensors are limited because of their resonant modes' low Q-factors by nature and additional attenuation caused by radiation loss. Furthermore, a great deal of biological/dielectric samples acquired in clinical practice are typically trace, and certain disease biomarker molecules are even incapable of depending on the nanoparticle-antibody system to amplify signals, making it even more challenging for low-Q metasurface biosensors to identify such samples. Thankfully, in recent years, the idea of quasi-bound state in the continuum (QBIC) has been incorporated into the design of metasurface based THz-dielectric sensors [125],[126].

Researchers using THz-waves for both time & frequency domain spectroscopy to analyze various materials as a result of recent advancements in THz-generation & detection techniques. The THz-dielectric sensor's modulation, can reveal information about the material on the sensing structure. The sensing mechanism of THz-dielectric sensors relies on how changes in the analyte's permittivity affect the excited SPP's propagation constant at the dielectric-metal surface boundary. 2D-materials such as SWNTs and graphene-based meta surfaces are highly suitable for the development of THz-dielectric sensors.

6. Terahertz refractive index sensors

THz- sensors uses sensing structures such as Si- photonic column arrays [124], graphene SPR [127], plasmonic wave guide [128], dielectric- graphene SPR [129], MMA [130], graphene- chiral meta surface [131], square SRR [132], graphene- photonic crystal [133], plasmonic ring resonator [134], EIT-like meta material [135], distributed Bragg reflectors [39], middle I-shaped metamaterial meta surface [136], E- shaped meta surface [137] and PIT- based graphene square ring [138] etc. In 2019, M. Nejat et al., [130] developed a MMA-based THz- sensor featuring SiO₂-wafer, MMA with a monolayer patterned graphene and a bottom Au- layer for the detection of RI of a sample. Q-factor, FoM and sensitivity of this sensor are determined by the analyte thickness and frequency of the THz- waves. This THz-RI-sensor is suitable for detecting glucose and malaria. In the same year, Elnaz Shokati et al., [131] reported a graphene chiral meta surface sensing structure for the detection of RI of samples. After 2 years, ZHONGGANG XIONG et al., [132] reported an innovative sensing structure called square SRR featuring a silicon substrate, gold reflective layer, poly ethylene dielectric layer and a metallic meta material that offered a sensitivity, Q- factor and FoM of 126 GHz/RIU, 44.17 and 10.5, respectively. The resonant frequency shift of this THz- sensor rises with rise in RI of the analyte. Also, sensitivity of the sensor improves with higher analyte thickness.

In 2022, Yuan-Kun Shi et al., [133] proposed a graphene- photonic crystal- based THz- sensor which works based on reflected Hännchenn-Goos Effect for the detection of RI of an analyte. Simultaneously, Ming Zhang et al., [139] reported an all- dielectric centre symmetry meta surface structure with identical cuboids based THz-sensor featuring GaAs- top player and Si-wafer. The frequency shift of this sensor is found to be rising with rise in analyte height. Sherin Thomas et al., [134] demonstrated that the sensitivity of THz- sensors used for RI sensing can be improved effectively by adopting a SIS (semiconductor-insulator-semiconductor) waveguide structure with an InSb- plasmonic ring resonator. Ring radius and RI of the sample and width of the wave guide shifts the rest and frequency of this sensor.

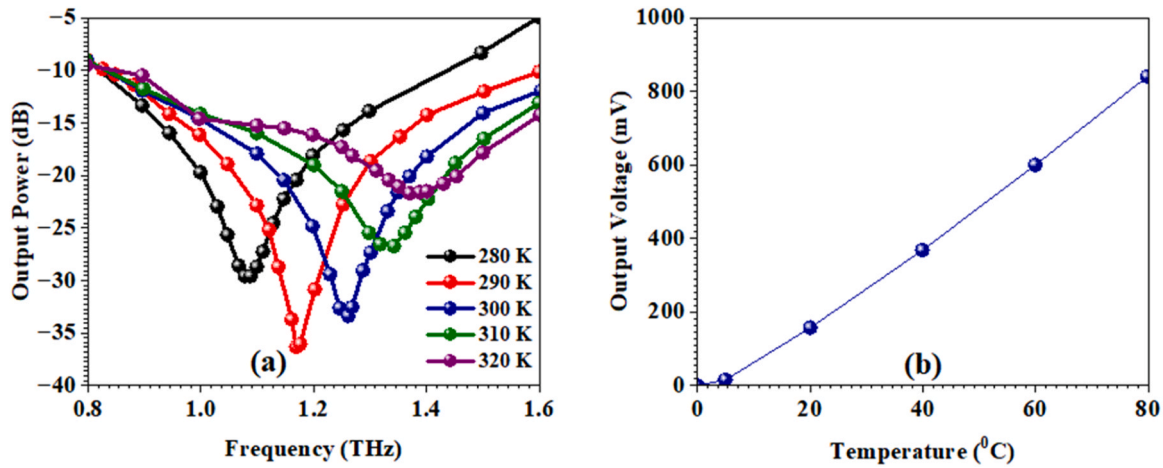


Fig. 13. (a) The output power of SPR waveguide based THz-temperature sensor with temperature variations at THz frequencies [151] (b) Output of CMOS integrated MMA-based uncooled THz-temperature sensor featuring a PTAT sensor [153].

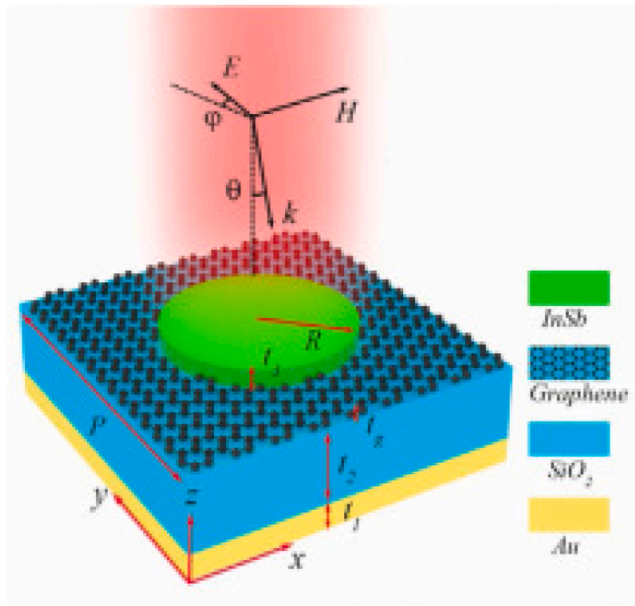


Fig. 14. Graphene MMA- based THz- temperature sensor (Reprinted from [154] with permission from Elsevier).

Pengfei Cao et al., [135] reported a EIT-like a meta material based THz-sensor featuring Si- wafer (high resistance) and structured aluminium layer. This sensor is a promising one for detecting glucose consideration. In 2023, Binghui Li et al., [136] reported a THz- sensor for the sensing of RI of samples featuring an electric SRR with T & I- shaped resonators on SOI-wafer (Fig. 10). The transmission spectra of this sensor with diverse RI values are depicted in Fig. 11. The resonant frequency decreases with rise in permittivity of analytes. Zhiqi Li et al., [140] reported a dielectric/graphene composite structured meta material based THz- sensor (Fig. 12) for the sensing of RI of analytes featuring 2 sensing layers and a TPX (three-period single layer of graphene/poly methyl pentene) and a prism. The sensing structures/mechanisms and performance of THz-sensors for the sensing of dielectrics & RI of analytes are listed in Table 4.

Si-photonic column arrays, SRRs, MMAs, graphene chiral meta surface square SRR, graphene-photonic crystal, plasmonic ring resonators, EIT-like meta material, distributed-Bragg reflectors, middle I-shaped metamaterial meta surface, E-shaped meta surface and PIT-based

graphene square ring etc are capable of sensing RI of samples. Graphene's 2D-structure allows for stronger confinement of SPP wave modes at THz-region in its surface, leading to enhanced optical-matter interactions compared to metal surfaces. Therefore, graphene has emerged recently as the most promising material for THz-sensors. Yet, there are still significant setbacks to plasmon excitation on graphene due to the comparatively high momentum mismatch between plasmons and optical photons. Plasmons on graphene have been effectively excited by means of periodically patterned structures of graphene such as micro/nano disks and micro/nano ribbons.

7. Terahertz sensors for temperature sensing

In the domains of biomedicine and material science, measuring and comprehending the thermal coefficient of chemical/ biological materials at THz frequency is crucial [149–159]. One can learn more about the heat transfer properties of various chemical/ biological compounds by measuring their THz thermal conductivities [149], [150]. In 2016, Jun Shibayama et al., [151] reported an SPR waveguide based THz-temperature sensor which is able to sense the temperature from 280 K to 320 K at 1.2 THz. The propagation aspects of the SPP waves are primarily responsible for the sensing performance. This sensor utilize SPP waves that are excited along the dielectric/metal interface. This THz-temperature sensor consisted of an InSb- layer deposited on the Si/SiO₂ waveguide. Water was used as an analyte. The output power of this THz sensor changes with temperature variations at THz frequencies (Fig. 13 (a)). With an elevation in temperature, the THz-temperature sensor's peak frequency rises. This is because, as temperature goes up, the real part of the permittivity of InSb considerably changes, going from -16.3 to -35.3 . Therefore, the primary cause of the peak frequency shift is thought to be a modification in the resonance state brought on by the change in InSb permittivity. In 2017, Youqiao Ma et al., [152] reported an integrated THz- temperature sensor featuring a Mach-Zehnder interferometer (MZI). This MZI can be obtained by placing a semiconductor layer onto a Si- membrane. The transmitted intensity modulates as a result of the interfering between the two SPP waves, which propagate over both faces of a semiconductor. The structure can be used as a temperature sensor because of its temperature-related transmission patterns. Fabry-Perot interferometer (FPI) and fibre Bragg grating (FBG) structures can also be used for measuring temperature of chemicals at THz-regime [20]. In 2020, Xu Wang et al., [153] reported a CMOS integrated MMA-based uncooled THz-temperature sensor featuring a PTAT sensor and triple band MMA (Fig. 13 (b)). This THz sensor was made out of a high-sensitive PTAT sensor and a highly absorbent MMA. It might be fully integrated into a conventional CMOS process,

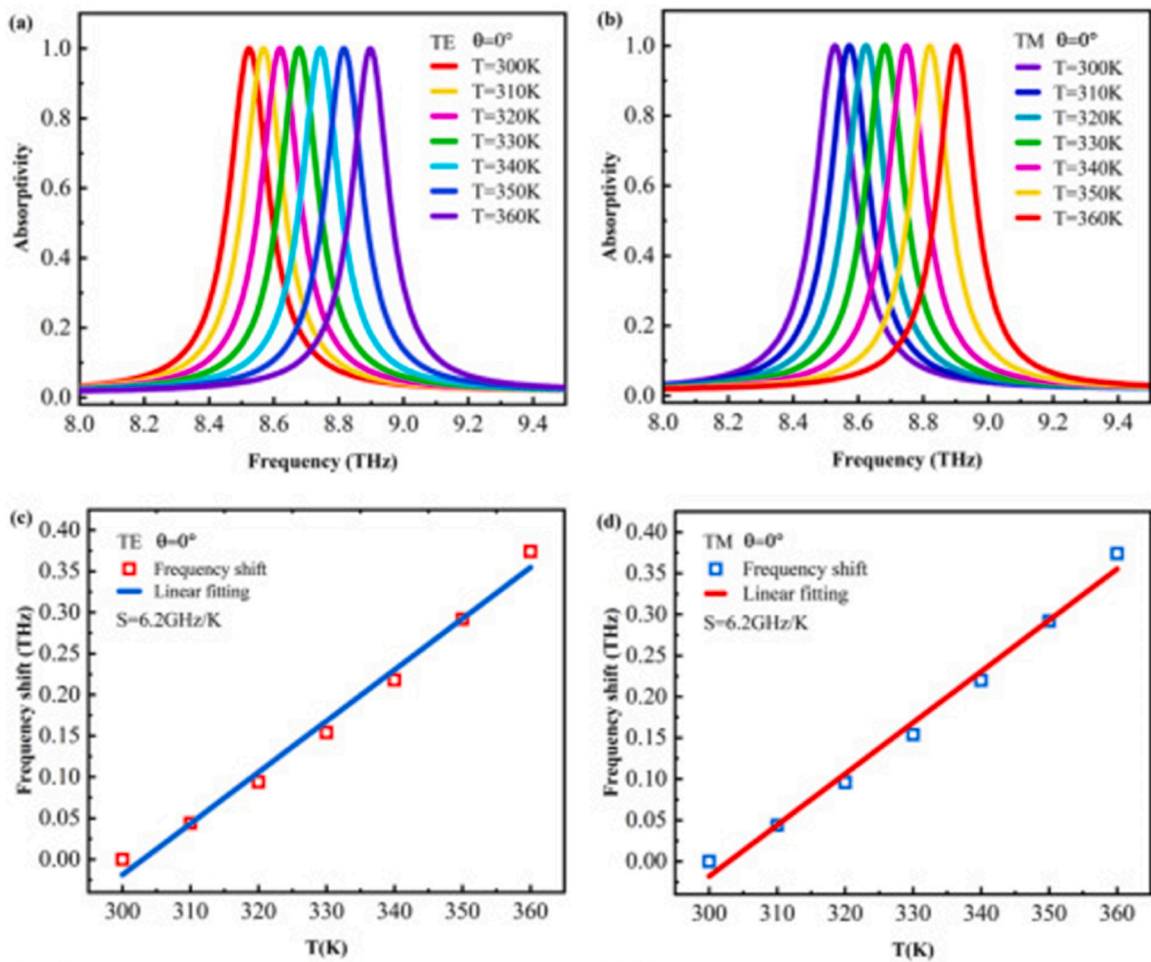


Fig. 15. Sensing characteristics of Graphene MMA- based THz- temperature sensor (Reprinted from [154] with permission from Elsevier).

Table 5

The sensing structures/mechanisms and performance of THz-temperature sensors.

Ref.	Sensing Structure/Mechanism	Sensing Range (K)	Sensitivity
[151]	SPR-Waveguide	280-320	7.5×10^{-3} THz/K
[152]	MZI-Waveguide	270-310	8.9×10^{-3} THz/K
[20]	FPI-Waveguide	323-353	-1.5 GHz/ $^\circ$ C
[20]	FBG-Waveguide	323-353	-1.32 GHz/ $^\circ$ C
[153]	MMA	273-353	10.11 mV/ $^\circ$ C
[145]	EIT-Meta material	310-350	2.32 GHz/K
[154]	Graphene MMA	300-360	6.2 GHz/K

significantly lowering the cost and complexity of manufacturing. The transducer gain of PTAT sensor was used to determine THz-sensors' temperature sensitivity. In 2023, Yi Ma et al., [145] reported a EIT-meta material based THz- temperature sensor featuring a T- shaped cut wire structure and a double SRR array. Polyimide was used as the wafer and gold film was used as the metal. The resonant frequency of this THz-temperature sensor changes with thermal variations. In the same year, Shilin Ma et al., [154] reported a graphene MMA- based THz-temperature sensor (Fig. 14) featuring Au-wafer, circle hole graphene, SiO₂- spacer and InSb- cylindrical spacer. The temperature sensing characteristics of this sensor is depicted in Fig. 15. The rise in temperature leads to the increase in frequency shift of the sensor and this sensor works with both TE & TM polarizations. The sensing

structures/mechanisms and performance of THz-temperature sensors are listed in Table 5.

8. Challenges in the development of THz-sensors

Label-free THz-sensors have intriguing applications in several industries including biotechnology, medical, ecology, pharmaceuticals and food quality control. This sort of sensing is simple and inexpensive to carry out. SPR-based THz-sensors have been the most suitable candidates for label-free sensing of biomolecules, chemicals, diseases and dielectrics. The THz-sensors are designed to optimize light-analyte interaction, resulting in high resolution & ultra high sensitivity sensing. However, THz-sensing has always struggled to match the sensitivity and resolution of optical sensing- due to its limited influence on natural materials. THz-SPPs (surface-plasmon-polaritons) have lower propagation/radiation loss than visible range SPPs, but have poor confinement on metal surfaces. Therefore, THz-SPPs are unsuitable for compact integration. This challenge can be overcome to certain limit by replacing metal surfaces with graphene strips. The MMA-THz-sensors excel at detecting changes in the MMAs-resonance frequency. MMAs have increased sensing versatility enabling for more sensitive and compact THz-sensors. The MMA-THz-sensors operates on the basis of changing reflection or transmission coefficients in accordance with the Q-factor. Number of MMAs have been utilized for THz-sensing applications. Due to the flaws in the design, real world applications were restricted. The biggest challenge is with the MMA's line width. The MMA's broad line width renders it inappropriate for real world

detection of diseases, chemicals and biomolecules. To enhance sensing capabilities, efficient MMAs with a narrow line width is essential. However, only limited MMAs were reported with such limited line width.

The dielectric or metallic metasurface based THz-sensors usually have relatively low sensitivity due to the naturally low Q-factor of their resonant modes. Moreover, these THz-sensors suffer from high radiation loss. Ultra thin metasurfaces are essential for suppressing undesirable losses due to the utilization of metal structures and response of resonant modes. The enhancement of electric field in sensing structures of THz-sensors is primarily focussed in the attachment of metal surfaces or split rings. Thus it is not possible to fully capitalize on the use of local electric field. Consequently, the next line of research will focus on refining the sensing structures and its sensitivity while taking into account the requirements of the real world applications, which include a simple sensing structure, ease of processing, manufacturing and testing. 2D-materials such as SWNTs and graphene can overcome these challenges. But the surface damaging of 2D-materials can directly influence the photoelectric behaviour of THz-sensors which is another serious concern in the development of THz-sensors.

With four unique advantages-(1) high Q-factor & high sensitivity, (2) minimum sample consumption, (3) label-free sensing without the need for additional reagents and (4) simple & precise detection and fast response-metamaterial based THz-sensors have enormous potential in the biomedical & other industrial fields. However, due to material bandgap constraints and expensive fabrication, THz-sensors have not yet reached their full potential for use in real world scenarios. SRRs can be used to obtain material properties. SRRs fabricated on very thin substrates (Examples: paper, Si_3N_4) offer good performance and high sensitivity. However, fabricating SRRs on thin substrates is a critical challenge. Assigning features noticed in the THz-spectra has proven to be one of the biggest challenges; the intricate inter-molecular interactions found in crystalline organic materials make it challenging to assign vibrational absorption bands. The lack of real time devices for generation & detection of THz waves, higher geometrical complexity of sensing structures and their integration & packaging are also challenges in the production of THz-sensors. Si-micromachining is known for its extremely high fabrication precision, but at THz frequencies, minute fabrication flaws could cause significant performance changes or degradation including frequency shifts, reduction in Q-factor, bandwidth changes and other unexpected losses. Therefore, high precise and quality fabrication techniques need to be developed for reaching their full potential for use in industrial applications.

9. Conclusion

THz-sensor development has accelerated recently, particularly in the fields of pharmacology, medicine, food industry, chemical industry, dielectric sensing, chemical sensing, temperature sensing and strain sensing. To analyse the minute variations in the RI, high sensitivity is very much essential. THz-sensors offer superior sensitivity to variations in ambient RI due to the unique distribution of resonance it excites. Graphene as a metamaterial is gaining tremendous popularity in the development of THz-sensors due to its unique benefits such as dynamic tunability and scalability. THz-sensors utilizes specific structures such as metamaterials, photonic crystals, periodic waveguide gratings, optical fibers, SRRs, MMAs, EIT/PIT-meta surfaces and dielectric/graphene composite structures for sensing dielectrics, RI, chemical/biological molecules, temperature & strain etc. Future research is required to investigate new materials and micro-nano architectures in order to substantially boost the sensitivity and also to enhance the scalability of THz-sensors for future biomedical and other industrial electronics applications.

CRedit authorship contribution statement

Ajayan J: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sreejith S.:** Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Manikandan M.:** Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Lai Wen-Cheng:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Saha Sumit:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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